

Qualibot-Hub — Project Task Board

Stack: MERN (MongoDB + Express + React + Node.js)

Docs: [qualiflow-hub/docs/](#) (Class Diagrams, Use Cases, Sequence Diagrams)

Reference: [docs/class_diagrams/](#) for all DB collections, [docs/sequence_diagrams/](#) for all API flows

Team Tracks

Track	Scope
[BE] Backend	Express API, Mongoose Models, Controllers, Services
[FE] Frontend	React Pages, Components, Axios Integration
[INF] Infrastructure	Auth, RBAC, Notifications Microservice, DevOps

Sprint 1 — Foundation

Goal: The project runs locally with a real DB, JWT auth works, and the tech stack is wired end-to-end.

ID	Task	Track	Difficulty	Depends On
BE-01	Initialize <code>server/</code> directory with Express, install deps <code>mongoose</code> , <code>express</code> , <code>jsonwebtoken</code> , <code>dotenv</code> , <code>cors</code> , <code>helmet</code>	[BE]	☐	—
BE-02	Connect MongoDB via Mongoose (<code>server/config/db.js</code>), add <code>.env</code> with <code>MONGO_URI</code> , <code>JWT_SECRET</code> , <code>PORT</code>	[BE]	☐	BE-01
BE-03	Create Mongoose models: <code>User</code> , <code>Site</code> , <code>Direction</code> , <code>Metier</code> (based on <code>db_01_identity_grh.puml</code>)	[BE]	☐☐	BE-02
BE-04	Create <code>POST /api/v1/auth/register</code> and <code>POST /api/v1/auth/login</code> endpoints with <code>bcrypt</code> + <code>JWT</code> generation	[BE]	☐☐	BE-03
INF-01	Write <code>server/middleware/auth.js</code> — <code>JWT</code> verification middleware (<code>verifyToken</code>)	[INF]	☐☐	BE-04
INF-02	Write <code>server/middleware/role.js</code> — <code>Global Role check</code> + Option B Processus isolation check (<code>checkRole</code> , <code>checkProcessus</code>)	[INF]	☐☐☐	INF-01
FE-01	Create <code>src/lib/api.ts</code> — <code>Axios</code> instance pointing to <code>http://localhost:5000/api/v1</code> , attach <code>JWT</code> from <code>localStorage</code> via interceptor	[FE]	☐	BE-04
FE-02	Replace all <code>Supabase</code> auth calls with new <code>Axios</code> endpoints in <code>src/hooks/useAuth.tsx</code>	[FE]	☐☐	FE-01
FE-03	Update <code>ProtectedRoute.tsx</code> to use new <code>JWT</code> auth state	[FE]	☐	FE-02

Sprint 2 — GRH & Core Models

Goal: Employee management and GRH module are fully operational with the DB.

ID	Task	Track	Difficulty	Depends On
BE-05	Create Mongoose models: <code>Employee</code> , <code>Fonction</code> , <code>Qualification</code> , <code>Habilitation</code>	[BE]	☐☐	BE-03

ID	Task	Track	Difficulty	Depends On
BE-06	Create Mongoose models: Formation, Formation_Demande, ValidationStepEmbed	[BE]	☐☐☐	BE-03
BE-07	Create WorkflowService — builds validation circuit chain based on Site.responsables_validation (See tech_seq_03)	[BE]	☐☐☐	BE-06
BE-08	CRUD routes for Employee (GET /employees, POST /employees, PUT /employees/:id)	[BE]	☐☐	BE-05
BE-09	Routes for Formation_Demande: POST /:id/submit, PATCH /:id/validate (circuit stepping)	[BE]	☐☐☐	BE-07, INF-02
BE-10	Create Processus model + CRUD routes. Expose GET /processus for all dropdowns across the app	[BE]	☐☐	BE-03
FE-04	Wire GRH page to real API (Employee list, search, add/edit) via api.ts	[FE]	☐☐	BE-08, FE-01
INF-03	Create Super Admin seeder script (server/scripts/seedSuperAdmin.js) to create the first user in DB	[INF]	☐	BE-04
INF-04	Build GET /users, POST /users/:id/role endpoints (Super Admin only, protected by INF-02)	[INF]	☐☐	INF-01, INF-02

Sprint 3 — Core QMS Modules

Goal: NC, Actions, Audits, KPI modules are functional with all cross-module triggers wired. This is the most critical sprint.

ID	Task	Track	Difficulty	Depends On
BE-11	Create Mongoose models: NC, Action, Sous_Action, Action_Log, Action_Parametre	[BE]	☐☐	BE-10
BE-12	Create ActionService — instancierAction(source, source_id, payload) shared service used by all modules (See tech_seq_01)	[BE]	☐☐☐	BE-11
BE-13	Full CRUD + Workflow for Action : submit, validate, update Sub-Action progress, auto-recompute taux_avancement, close (See tech_seq_04)	[BE]	☐☐☐☐	BE-12, INF-02
BE-14	Create Mongoose models: Audit, Audit_Parametre, Constat_Ecart, ChecklistISO	[BE]	☐☐	BE-10
BE-15	Full Audit workflow routes: plan, checklist, declare ecart → auto-create NC + Action (See tech_seq_01)	[BE]	☐☐☐☐	BE-12, BE-14
BE-16	Create Mongoose models: Indicateur, Indicateur_Valeur, Historique_Cible, Axe_Politique, Indicateur_Type	[BE]	☐☐	BE-10
BE-17	KPI routes: create indicateur, add period value, check threshold → auto-create NC if below seuil_nc, MongoDB aggregate pipeline for dashboard (See tech_seq_05)	[BE]	☐☐☐	BE-12, BE-16
FE-05	Wire Actions page: form submission, validation circuit status, sub-action progress tracker	[FE]	☐☐☐	BE-13
FE-06	Wire Audits page: plan form, checklist UI, ecart declaration modal	[FE]	☐☐☐	BE-15
FE-07	Wire KPI page: indicator setup, period value input, chart rendering (Recharts)	[FE]	☐☐☐	BE-17

Sprint 4 — Secondary Modules

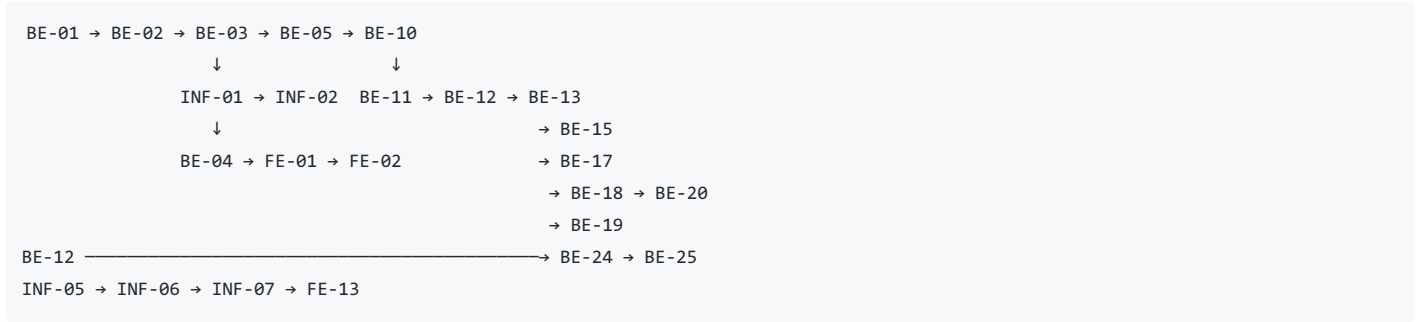
Goal: PNC, Suppliers, Clients, Metrology, Risks, Documentation fully operational.				
ID	Task	Track	Difficulty	Depends On
BE-18	Models + routes for PNC : declare fiche, add detail items, decision + validation circuit, close (See tech_seq_02)	[BE]	🔴🔴🔴	BE-12
BE-19	Models + routes for Fournisseur : CRUD, evaluate (weighted score calc), reclamation → link to PNC (See tech_seq_07)	[BE]	🔴🔴🔴	BE-12
BE-20	Models + routes for Client : CRUD, reclamation (with with_return → PNC trigger), questionnaire templates, launch enquete (UUID link), public POST /public/enquetes/:lien/reponse (See tech_seq_08)	[BE]	🔴🔴🔴🔴	BE-18
BE-21	Models + routes for Equipement_Mesure : interventions (auto-suspend + action if NC), R&R analysis calculation (See tech_seq_06)	[BE]	🔴🔴🔴	BE-12
BE-22	Models + routes for Risque_Fiche : initial score (formula eval), threshold → auto-create Action, residual score update (See tech_seq_09)	[BE]	🔴🔴🔴	BE-12
BE-23	Models + routes for DocumentFile : upload (S3/blob), 5-axis structure per Processus, verification circuit, diffusion, accusé de réception tracking (See tech_seq_10)	[BE]	🔴🔴🔴🔴	BE-10
FE-08	Wire PNC page: declare form, product detail list, treatment decision workflow	[FE]	🔴🔴🔴	BE-18
FE-09	Wire Fournisseurs & Clients pages	[FE]	🔴🔴	BE-19 , BE-20
FE-10	Wire Métrologie and Risques pages	[FE]	🔴🔴	BE-21 , BE-22
FE-11	Restructure Documentation page with 5-axis tab layout per Processus	[FE]	🔴🔴🔴	BE-23

Sprint 5 — Reviews, RBAC & Infrastructure (Semaine 7–8)

Goal: Revues, Notifications microservice, and final RBAC polishing.				
ID	Task	Track	Difficulty	Depends On
BE-24	Models + routes for RevueProcessus : GET /generate (parallel MongoDB aggregations), POST /save (explode constats → Actions) (See tech_seq_11)	[BE]	🔴🔴🔴🔴	BE-17 , BE-15 , BE-13
BE-25	Models + routes for RevueDirection : same aggregation pattern from multiple modules into one summarized report	[BE]	🔴🔴🔴🔴	BE-24
BE-26	Models + routes for Reunion (CRUD, Participants, Decisions → auto-link Actions)	[BE]	🔴🔴	BE-13
BE-27	Models + routes for VeilleItem (CRUD, Compliance Status → force Action creation), PlanningEvent (4 flux)	[BE]	🔴🔴	BE-13
INF-05	Initialize the Notification Microservice as a separate Node.js process (server/notification-service/) (See tech_seq_13)	[INF]	🔴🔴🔴	BE-01
INF-06	Configure Redis Pub/Sub (or RabbitMQ) broker; emit events from Action , PNC , Audit routes	[INF]	🔴🔴🔴🔴	INF-05
INF-07	Implement Socket.IO server on the Microservice; connect to React frontend (in-app bell notifications)	[INF]	🔴🔴🔴	INF-06
INF-	Implement Nodemailer HTML email templates for ACTION_CREATED , PNC_DECLARED , AUDIT_PLANNED			INF-06

ID	events Task	[INF] Track	🔥 Difficulty	Depends On
FE-12	Build pages for Revue de Processus & Revue de Direction (auto-fill + manual text sections)	[FE]	🔥🔥	BE-24 , BE-25
FE-13	Build Notification Bell UI in Navbar + Notification History page (Socket.IO consumer)	[FE]	🔥🔥	INF-07
FE-14	UI enforces role restrictions: hide action buttons for viewer , disable cross-processus edits for editor	[FE]	🔥	INF-02

Dependency Graph (Critical Path)



Legend

Symbol	Meaning
☐	Easy (< 2h)
☐☐	Medium (2–4h)
☐☐☐	Hard (4–8h)
☐☐☐☐	Very Hard (1–2 days)
[BE]	Backend developer task
[FE]	Frontend developer task
[INF]	Infrastructure / DevOps task